

National University of Computer and Emerging Sciences
Lahore Campus

**Machine Learning for Robotics
(CS4090)**

Date: April 8th 2026

Course Instructor

Ms. Sarah Khaleel

Sessional-II Exam

Total Time: 1 Hour

Total Marks: 32

Total Questions:

Roll No

Section

Student Signature

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- Instruction/Notes:**
- Attempt all questions on the provided separate answer sheet.
 - Clearly write corresponding question number and part number at the top center of the answer sheet with a thick pen / marker before starting a new question / answer.
 - Show all steps clearly.
 - Write the formula before substitution.
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CLO 1 (Q1 to Q2): Demonstrate basic understanding of Machine Learning Algorithms and optimization methods.

[5+22+5=32 Marks]

Q1: Choose the correct option. (1*5=5 Marks)

- i) When an experienced deep learning engineer works on a new problem, they can usually use insight from previous problems to train a good model on the first try, without needing to iterate multiple times through different models. True/False?
- a. True
b. False

Ans: b

- ii) Logistic regression's weights w should be initialized randomly rather than to all zeros, because if you initialize to all zeros, then logistic regression will fail to learn a useful decision boundary because it will fail to "break symmetry", True/False?
- a. True
b. False

Ans: Logistic Regression doesn't have a hidden layer. If you initialize the weights to zeros, the first example x fed in the logistic regression will output zero but the derivatives of the Logistic Regression depend on the input x (because there's no hidden layer) which is not zero. So at the

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second iteration, the weights values follow x 's distribution and are different from each other if x is not a constant vector.

iii. What does a neuron compute?

- a. A neuron computes an activation function followed by a linear function ($z = Wx + b$)
- b. A neuron computes a linear function ($z = Wx + b$) followed by an activation function
- c. A neuron computes a function g that scales the input x linearly ($Wx + b$)
- d. A neuron computes the mean of all features before applying the output to an activation function

Ans: b

iv. Which of the following are true?

- 1. X is a matrix in which each column is one training example.
 - 2. $a^{[2]}_4$ is the activation output by the 4th neuron of the 2nd layer
 - 3. $a^{[2]}(12)$ denotes the activation vector of the 2nd layer for the 12th training example.
 - 4. $a^{[2]}$ denotes the activation vector of the 2nd layer.
- a. 1 b. 2 and 4 c. 3 d. All of the above

Ans: d

- v. Suppose you have built a neural network. You decide to initialize the weights and biases to be zero. Which of the following statement is True?
- a. Each neuron in the first hidden layer will perform the same computation. So even after multiple iterations of gradient descent each neuron in the layer will be computing the same thing as other neurons.
 - b. Each neuron in the first hidden layer will perform the same computation in the first iteration. But after one iteration of gradient descent they will learn to compute different things because we have "broken symmetry".
 - c. Each neuron in the first hidden layer will compute the same thing, but neurons in different layers will compute different things, thus we have accomplished "symmetry breaking" as described in lecture.
 - d. The first hidden layer's neurons will perform different computations from each other even in the first iteration; their parameters will thus keep evolving in their own way.

Ans: a

Q2: (2*5+3*4=22 Marks)

i) Consider the following code:

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```
A = np.random.randn(4,3)
B = np.sum(A, axis = 1, keepdims = True)
What will be B.shape?
```

Ans: B.shape = (4, 1)

we use (keepdims = True) to make sure that A.shape is (4,1) and not (4,). It makes our code more rigorous.

ii) Consider the two following random arrays "a" and "b":

```
a = np.random.randn(2, 3) # a.shape = (2, 3)
```

```
b = np.random.randn(2, 1) # b.shape = (2, 1)
```

```
c = a + b
```

What will be the shape of "c"?

Ans: b (column vector) is copied 3 times so that it can be summed to each column of a. Therefore, c.shape = (2, 3).

iii) Consider the two following random arrays "a" and "b":

```
a = np.random.randn(4, 3) # a.shape = (4, 3)
```

```
b = np.random.randn(3, 2) # b.shape = (3, 2)
```

```
c = a * b
```

What will be the shape of "c"?

Ans: "*" operator indicates element-wise multiplication. Element-wise multiplication requires same dimension between two matrices. It's going to be an error.

iv) Suppose you have n_x input features per example. Recall that $X=[x^{(1)}, x^{(2)}...x^{(m)}]$. What is the dimension of X?

Ans: (n_x , m)

v) Consider the following code snippet:

```
# a.shape = (3,4)
```

```
# b.shape = (4,1)
```

```
for i in range(3):
```

```
    for j in range(4):
```

```
        c[i][j] = a[i][j] + b[j]
```

How do you vectorize this?

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Ans: $c = a + b.T$

- vi) Recall that `np.dot(a,b)` performs a matrix multiplication on `a` and `b`, whereas `a*b` performs an element-wise multiplication.

Consider the two following random arrays "a" and "b":

```
a = np.random.randn(12288, 150) # a.shape = (12288, 150)
b = np.random.randn(150, 45) # b.shape = (150, 45)
c = np.dot(a, b)
```

What is the shape of `c`?

Ans: $c.shape = (12288, 45)$, this is a simple matrix multiplication example.

- vii) Consider the following code:

```
a = np.random.randn(3, 3)
b = np.random.randn(3, 1)
c = a * b
```

What will be `c`?

Ans: This will invoke broadcasting, so `b` is copied three times to become (3,3), and `*` is an element-wise product so $c.shape = (3, 3)$.

- viii) Write the correct vectorized implementation of forward propagation for layer l , where $1 \leq l \leq L$?

**Ans: $Z^{[l]} = W^{[l]}A^{[l-1]} + b^{[l]}$
 $A^{[l]} = g^{[l]}(Z^{[l]})$**

- ix) Suppose `img` is a (32,32,3) array, representing a 32x32 image with 3 color channels red, green and blue. How do you reshape this into a column vector?

Ans: `x = img.reshape((32 * 32 * 3, 1))`

Question 3: (5 Marks)

Consider the following 1 hidden layer neural network:

`b[1]` will have shape (4, 1)
`W[1]` will have shape (4, 2)
`W[2]` will have shape (1, 4)
`b[2]` will have shape (1, 1)

Draw the computation graph and mention the required computations along with the dimensions for `z1`, `z2`, `a1`, and `a2`.

Ans:

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